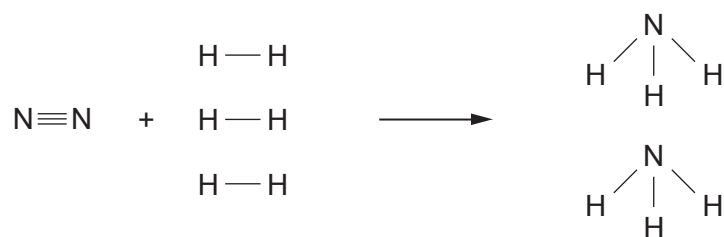


7. (a) The formation of ammonia can be represented by the following equation.



- The total amount of energy released in making the bonds in the products is 2340 kJ
- The total amount of energy released in making the bonds in the products is 94 kJ **more than** the total energy used in breaking the bonds in the reactants
- The amount of energy used to break the $\text{N}\equiv\text{N}$ bond is 941 kJ

Use this information to calculate the energy used to break **one** $\text{H}-\text{H}$ bond.

[3]

Energy = kJ



- (b) The Haber process used to make ammonia is usually carried out at a temperature of around 400°C . Explain why this is the optimum temperature.

[2]

A higher temperature is not used because

.....

A lower temperature is not used because

.....

- (c) Ammonia is used in the production of fertilisers such as ammonium nitrate.

Give the balanced symbol equation for the reaction between nitric acid and ammonia to form ammonium nitrate.

[2]

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- (d) Eutrophication is a problem caused by fertilisers being washed into waterways leading to overgrowth of plants. Describe how eutrophication leads to the death of aquatic organisms.

[2]

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